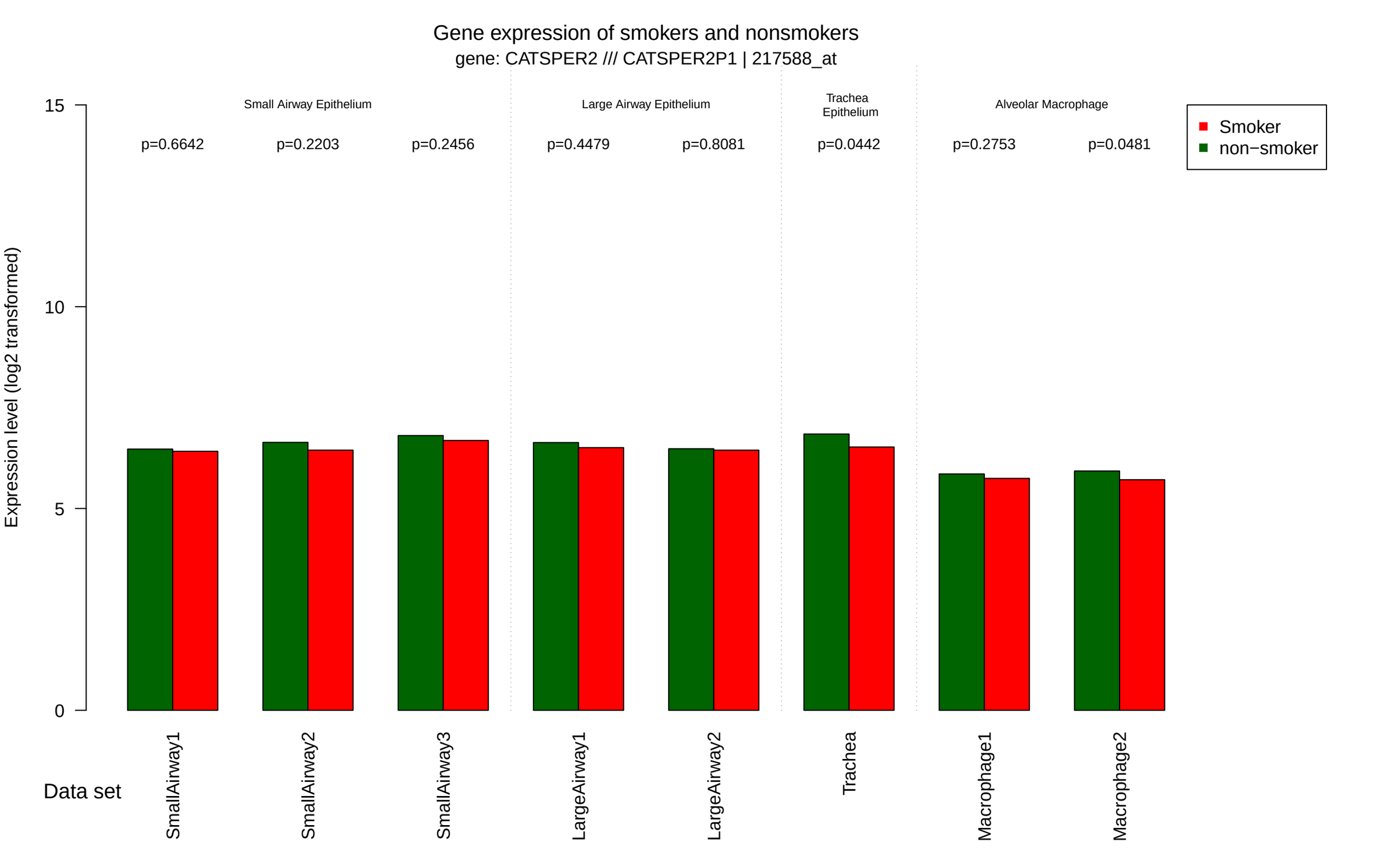
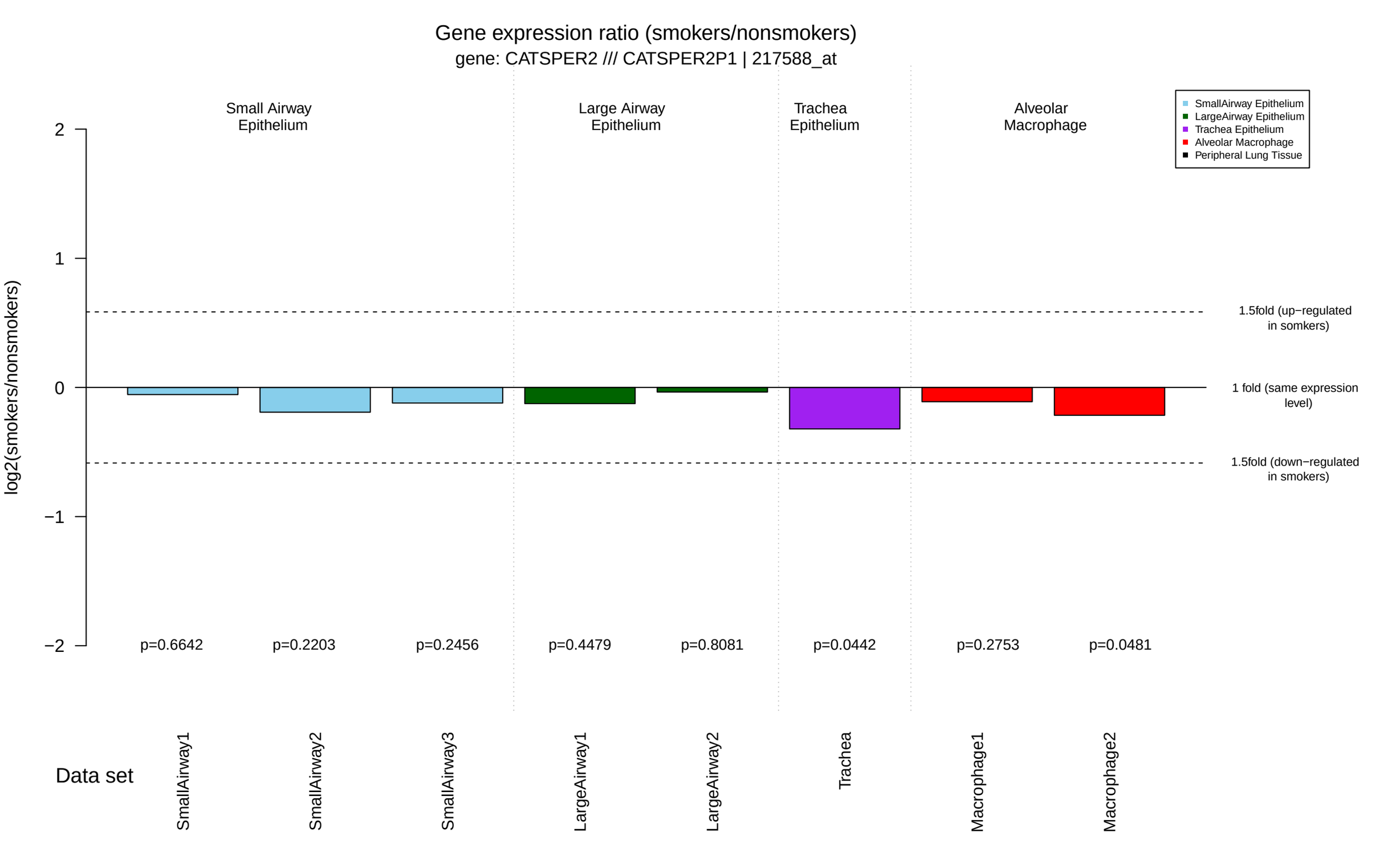
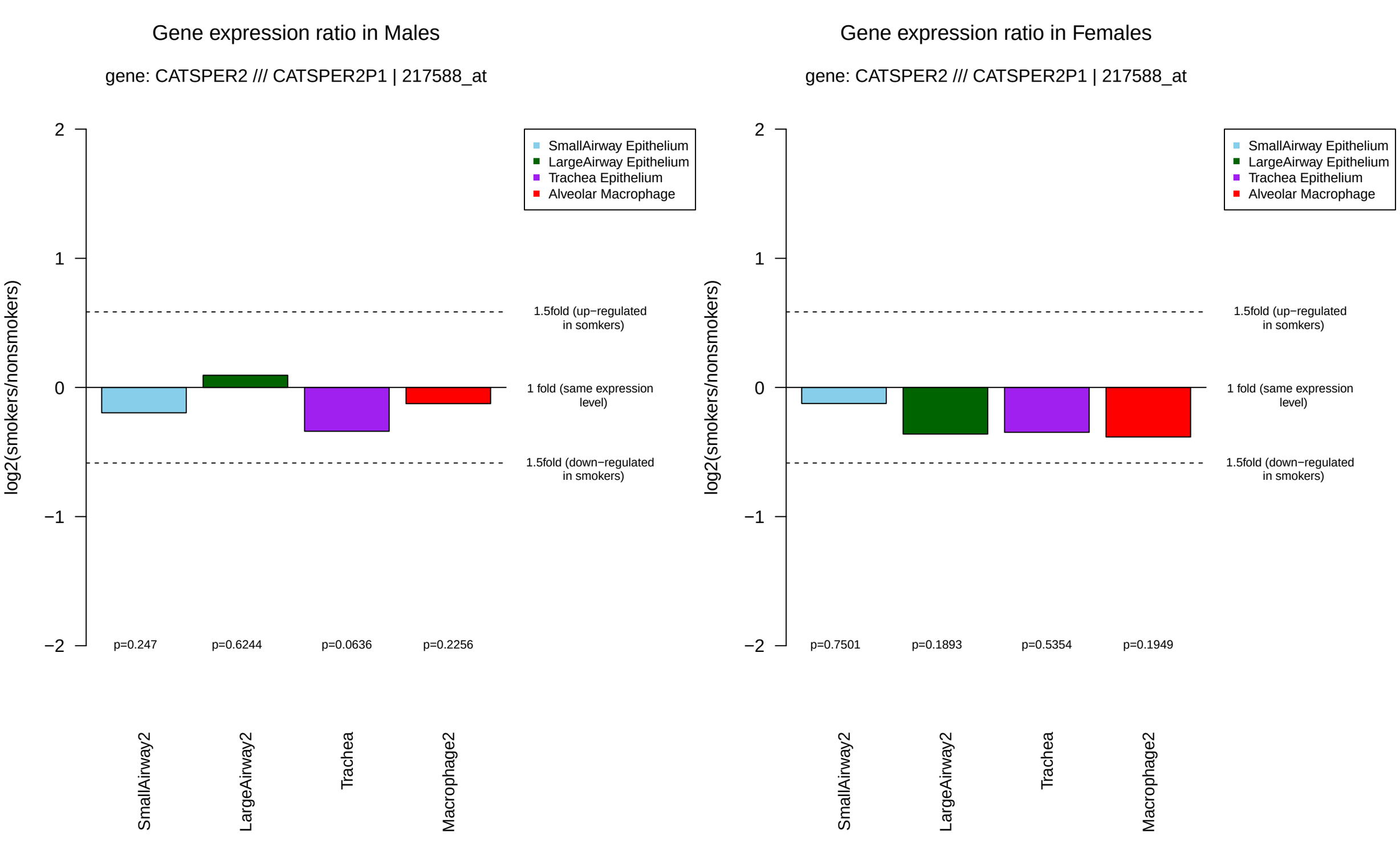
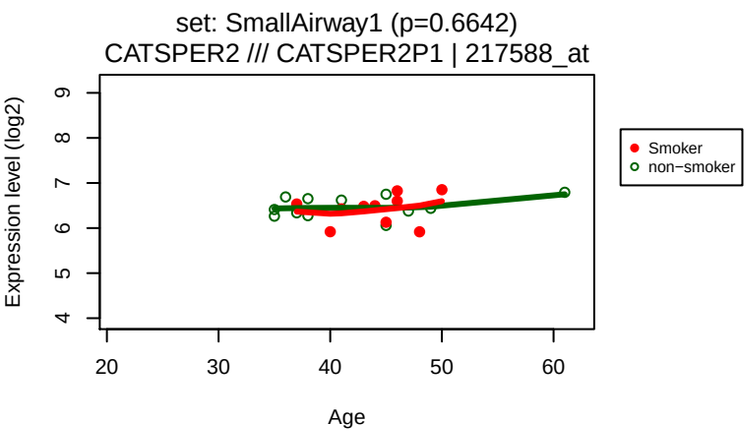
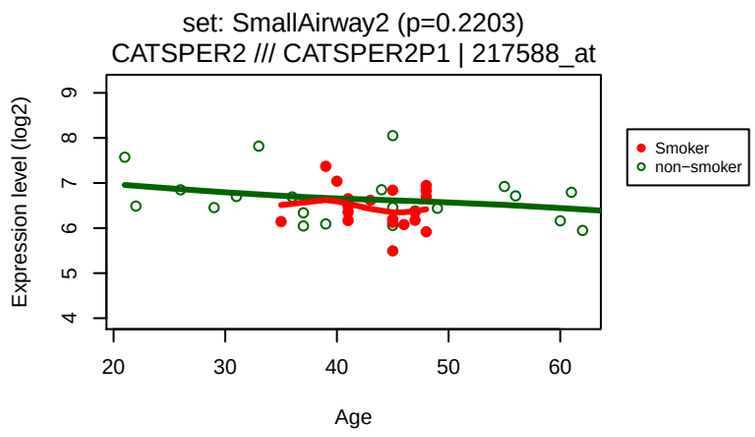
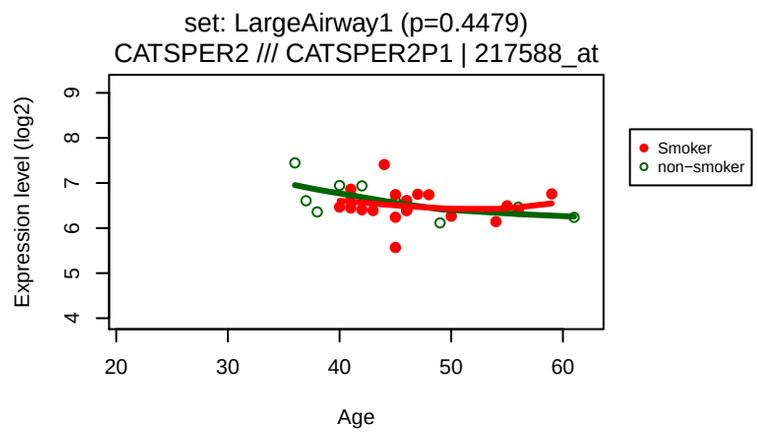

Gene expression of smokers and nonsmokers
gene: CATSPER2 /// CATSPER2P1 | 217588_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6642
p=0.2203
p=0.2456
p=0.4479
p=0.8081
p=0.0442
p=0.2753
p=0.0481
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

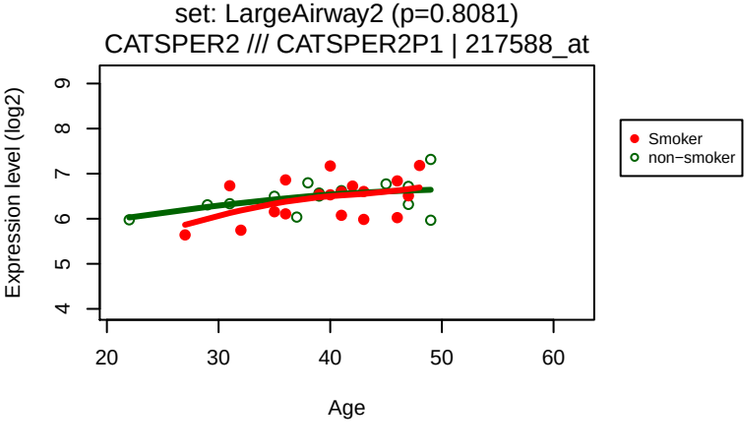

Gene expression ratio (smokers/nonsmokers)
gene: CATSPER2 /// CATSPER2P1 | 217588_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6642
p=0.2203
p=0.2456
p=0.4479
p=0.8081
p=0.0442
p=0.2753
p=0.0481
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

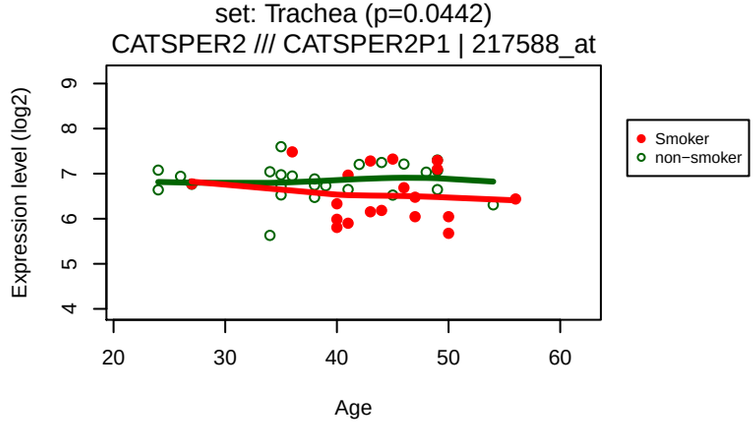

Gene expression ratio in Males
gene: CATSPER2 /// CATSPER2P1 | 217588_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.247
p=0.6244
p=0.0636
p=0.2256
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CATSPER2 /// CATSPER2P1 | 217588_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7501
p=0.1893
p=0.5354
p=0.1949
SmallAirway2
LargeAirway2
Trachea
Macrophage2

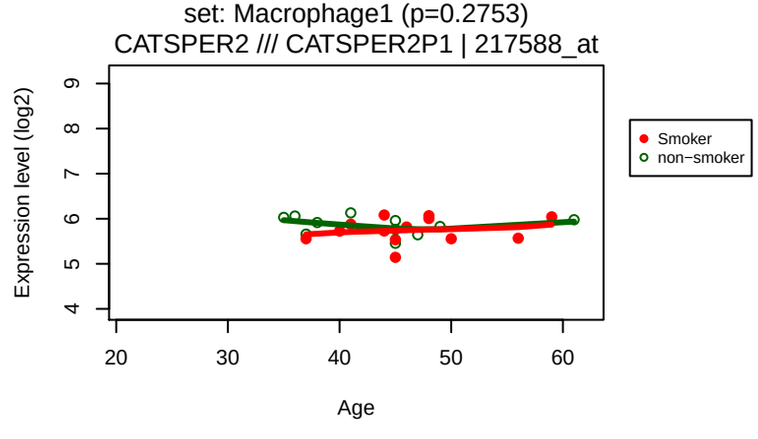

set: SmallAirway1 (p=0.6642)
CATSPER2 /// CATSPER2P1 | 217588_at
Expression level (log2)
Age
Smoker
non−smoker

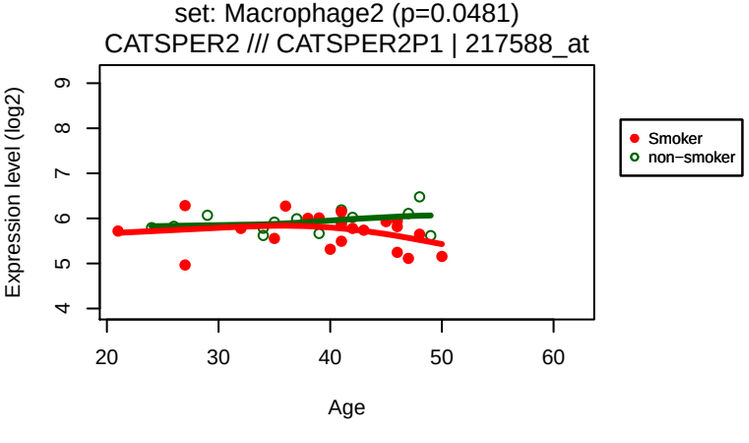

set: SmallAirway2 (p=0.2203)
CATSPER2 /// CATSPER2P1 | 217588_at
Expression level (log2)
Age
Smoker
non−smoker

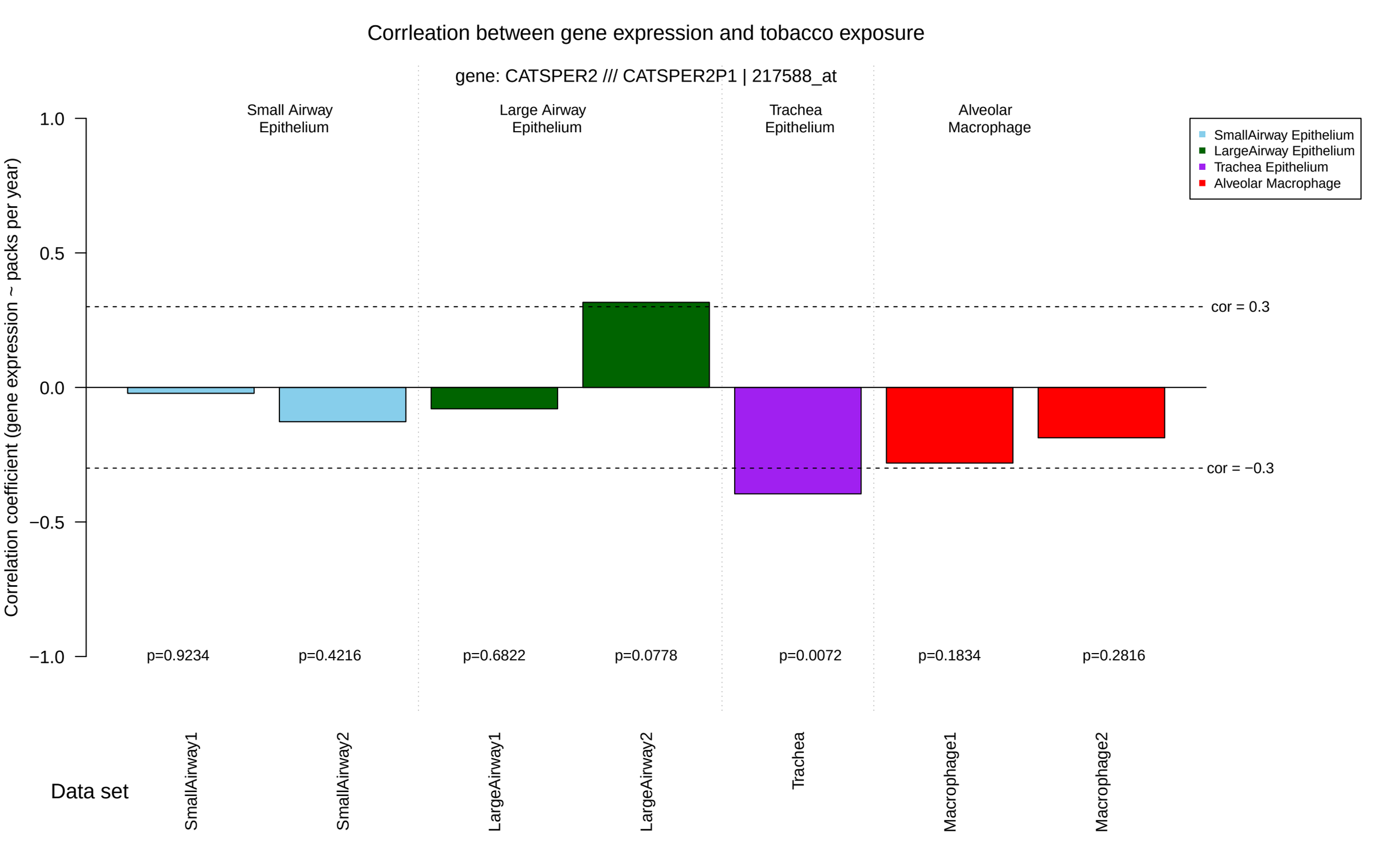

set: LargeAirway1 (p=0.4479)
CATSPER2 /// CATSPER2P1 | 217588_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8081)
CATSPER2 /// CATSPER2P1 | 217588_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0442)
CATSPER2 /// CATSPER2P1 | 217588_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2753)
CATSPER2 /// CATSPER2P1 | 217588_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0481)
CATSPER2 /// CATSPER2P1 | 217588_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CATSPER2 /// CATSPER2P1 | 217588_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9234
p=0.4216
p=0.6822
p=0.0778
p=0.0072
p=0.1834
p=0.2816
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2